

EE2080: Random Processes - Final Exam

Maximum Marks: (70 + 10*)

Time Allowed: 2.5 hours

04 Sept 2025

This is a closed-book exam. If an electronic gadget (anything with memory or that can connect to the internet) is found on your person, it will be treated as resorting to unfair means regardless of whether or not you were using the device. Please mention your name and roll number clearly in the answer script. ALL THE BEST!!!

Mean, Correlation and Stationarity (10 marks)

Q1: Consider the four functions shown in Figure 1. Which of these are valid autocorrelation functions? Justify. (2 marks)

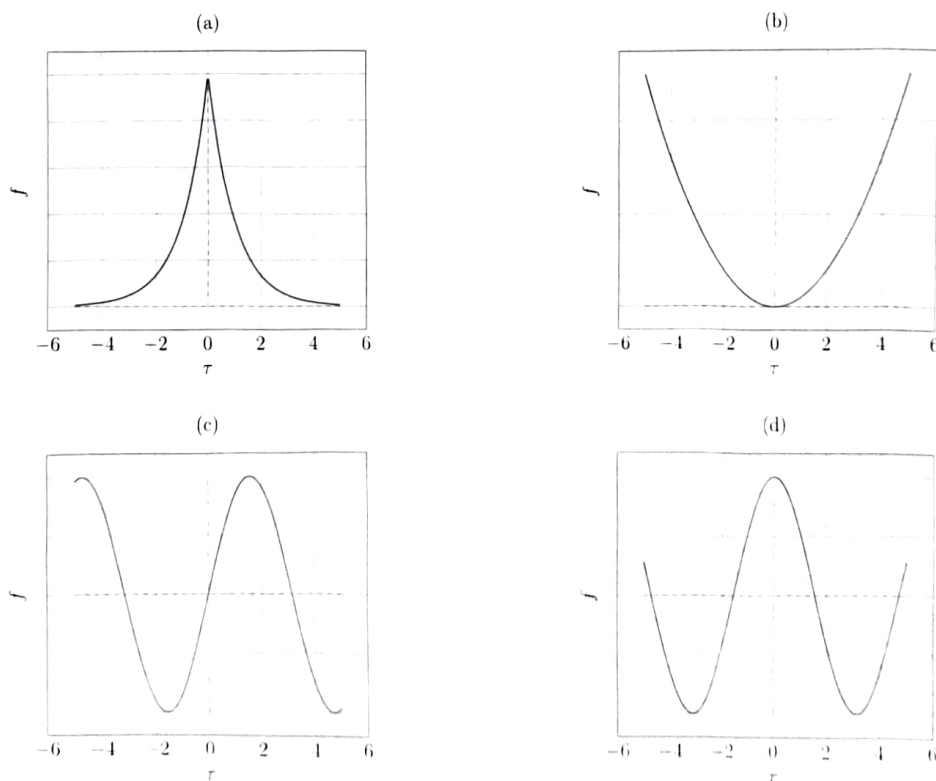


Figure 1: **Q1:** Which of these functions are valid autocorrelation functions?

Q2: Let A be a nonnegative random variable with $\mathbb{E}[A^2] < \infty$, and let $\Theta \sim \text{Unif}[0, 2\pi)$, independent of A . Define the continuous-parameter random process $Y_t = A \cos(\omega_0 t + \Theta)$, $t \in \mathbb{R}$.

(a) Is $\{Y_t, t \in \mathbb{R}\}$ wide-sense stationary? Justify. (2 marks)

(b) Is $\{Y_t, t \in \mathbb{R}\}$ ergodic in mean? Is it ergodic in autocorrelation? Justify your answers. (2 marks)

Q3: Consider a discrete parameter stochastic process $\{X_n, n \geq 0\}$, which has both the stationary increment and independent increment properties. Also, assume that $X_0 = 0$. Prove that

(a) The mean and variance of X_n are linear in n . (2 marks)

(b) For $l \geq k$, the covariance $C_X(k, l) = \text{var}(X_k)$. (1 mark)

(c) Can such an X_n process be stationary? Justify. Give corner cases, if any, for your answer. (1 mark)

Poisson Processes

(15 marks)

Q4: Let a counting process $\{N_t, t > 0\}$ be a Poisson process with rate λ . For $0 < s < t$, show that given $N_t = n$, N_s is a binomial random variable with parameters n and $\frac{s}{t}$. (5 marks)

Q5: Let a counting process $\{N_t, t > 0\}$ has the stationary and independent properties and for all $t > 0$, has the pmf $\mathbf{p}_{N_t}(\mathbf{n}) = \frac{(\lambda t)^n \exp(-\lambda t)}{n!}$, $n = 0, 1, 2, \dots$. Let X_1 be the epoch of the first arrival and X_n be the interarrival time between the $(n-1)^{\text{th}}$ and the n^{th} arrivals. Using only these assumptions, prove the following:

• $Pr(X_1 > x) = \exp(-\lambda x)$ (2 mark)

• Let S_{n-1} be the epoch of the $(n-1)^{\text{th}}$ arrival. Show that $Pr(X_n > x | S_{n-1} = \tau) = \exp(-\lambda x)$. (3 mark)

• For each $n > 1$, show that $Pr(X_n > x) = \exp(-\lambda x)$ and that X_n is independent of S_{n-1} . (3 mark)

• Argue that the inter-arrival times are i.i.d exponential. (2 mark)

Markov Chains

(40 + 10* marks)

Q6: Let $\{X_n, n \geq 0\}$ be a discrete time Markov chain with the transition probability diagram given in Figure 2. For this DTMC,

(a) Write the transition probability matrix P . (1 mark)

(b) Assuming the initial PMF to be discrete uniform on the state space, give the PMF of X_2 . (2 mark)

(c) Find the first recurrence time distributions for all the four states. (4 marks)

(d) For each $i \in \{0, 1, 2, 3\}$, find the probabilities $f_{i,i}$. (2 marks)

(e) Classify the states as transient or recurrent based on (d) above. (2 mark)

(f) If there are recurrent states, compute their mean recurrence times. (2 marks)

Handwritten calculations on the right margin:

$$\begin{array}{r} 2 \\ 35 \\ 140 \\ 35 \times 4 = 140 \\ 490 \\ 490 \\ 225 \\ 225 \times 2 = 450 \\ 400 \end{array}$$

$$\begin{array}{r} 3 \\ 16 \\ 48 \\ 96 \\ 72 \\ 111 \\ 111 \\ 111 \\ 400 \end{array}$$

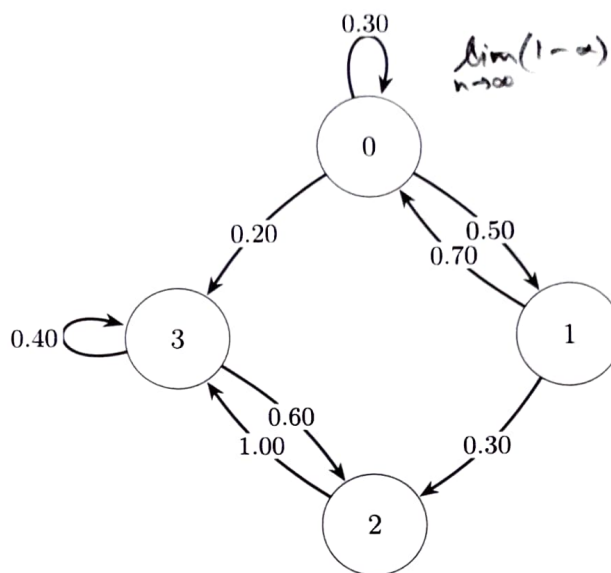


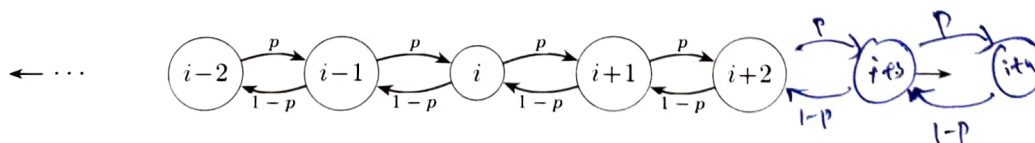
Figure 2: **Q6:** Transition Probability Diagram

- (g) Based on (f) above, classify the states as positive or null recurrent. (1 mark)
- (h) Find the stationary distribution. (3 marks)
- (i) If there are positive recurrent states in the chain, relate their mean recurrence times with their steady state probabilities. (1 mark)
- (j) Is the DTMC aperiodic? Justify. (2 marks)

Q7: Consider a DTMC $\{X_n, n \geq 0\}$ with the state space $S = \{0, 1, 2, 3, \dots\}$ and the transition probability matrix P . For all $i \in S$, let $P_{i,j} = \frac{1}{2^{j+1}}$, $j \in S$. Prove that $X_1, X_2, \dots$ is a sequence of independent and identically distributed random variables. (5 marks)

Q8: Consider a random walk on $\mathbb{Z}$ whose transition probability diagram is given below. Derive a condition for transience or recurrence of the states, i.e., for which values of p is the chain transient/recurrent. (5 marks)

(Hint: Use Lavrov's Lemma: $\frac{4^n}{2n+1} < \binom{2n}{n} < 4^n$, or any other valid approximation of binomial coefficients of your choice.)



Random walk on $\mathbb{Z}$: $P(i \rightarrow i+1) = p$, $P(i \rightarrow i-1) = 1-p$.

Q9: Consider a router with infinite packet storage, called an unbounded queue. At every time step, with probability $p = \frac{1}{3}$, one packet arrives at the router, and independently, with probability $q = \frac{1}{2}$, one packet leaves the router. Note that at any particular time instant, there can be both an arrival and a departure, either one of these or neither. When the queue is empty, there can only be an arrival or no arrival, i.e., packet departure is not a valid event. In the

steady state, what is the average number of packets in the queue? What is the mean time for the queue to get empty? **(10 marks)**

(**Hint:** Using an initial set of 2-3 global balance equations, hypothesize an expression for the stationary probability π_i in terms of π_0 , verify that the hypothesis satisfies all the global balance equations, use the normalization property to find π_0 , then find the required quantities using the obtained stationary distribution.)

Q10: (Bonus Question) Consider an 8×8 standard chess board and a knight's tour on it (If you are unfamiliar with the rules of chess, the allowed moves of a knight are shown in the figure below). At any time step, the knight selects one of the next permissible positions uniformly at random independently of the past.

(a) Justify why the knight's tour is a DTMC. What is the state space? Is the DTMC irreducible? Is it aperiodic? If not, what is the period? Justify. **(3 marks)**

(b) Find the stationary distribution of the DTMC. **(7 marks)**

(**Hint:** There is no chance of solving a set of 64 linear equations in 64 unknowns, make use of symmetry from the centre of the board, the squares at the 4 corners of the board must have the same probability under the stationary distribution, further a square at the centre of the board has a higher probability than one at the corner (Justify).)

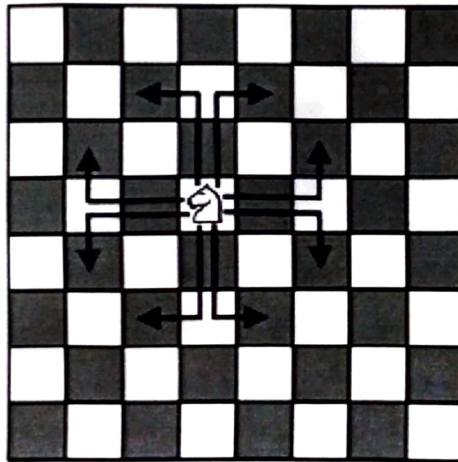


Figure 3: Permissible moves of a knight

*Write your thought process for this question even if you are unable to arrive at a final solution.

$$\begin{array}{r} 4 \\ 27 \\ \times 7 \\ \hline 189 \end{array}$$